TECHNICAL USER MANUAL

FWD3D

Forward modeling for gravity, magnetic, vector-magnetic, and DIGHEM-style AEM data

MATLAB implementation in the Joint3D repository

Primary script: bin\FWD3D.m

Sample input: example1\FWD3D\inpt3.m

Manual scope: Configuration, execution, outputs, visualization, and troubleshooting

Source review: 23 July 2026

Repository-specific manual based on the distributed source and example artifacts

Contents and reading path

Start with Quick start if you want to reproduce the supplied example. Use the input-reference sections when building a new model, and consult Outputs and troubleshooting when integrating the synthetic data with INV3D.

> **Important.** FWD3D is a script, not a function. It takes no arguments, runs the file named inpt3.m, and writes all top-level results under MATLAB's current working folder.

1. Purpose and operating model

FWD3D builds synthetic geophysical observations for up to four fixed physics slots: gravity, induced magnetics, vector magnetic response, and frequency-domain airborne electromagnetics (AEM). It also creates field maps and model images. The output data files are directly usable as synthetic observation inputs for the companion INV3D workflow.

What the script does

- Runs inpt3.m to place all configuration variables in the MATLAB workspace.
- Loops over the four property slots and skips any slot disabled by compFlag.
- Builds the receiver/channel table, model grid, background model, and total property model.

- Calls the physics-specific forward solver.
- Writes obsDataN.dat, field-map PNG files, and model-view PNG files for each enabled slot.
- For AEM, creates per-channel solver work folders under work.

Execution flow

```
inpt3.m
   -> compFlag selects physics slots
   -> receiver/channel table (getBulk, getRpars)
   -> model grid and property model (getGpars, getMpars)
   -> forward solver (gravity / magnetic / vector magnetic / AEM)
   -> obsDataN.dat + FieldsPN/*.png + ModelPN/*.png
```

2. Quick start

The safest launch pattern is to put bin and all of its subfolders on the MATLAB path, then change to the example working folder before calling FWD3D.

```
repoRoot = 'D:\UofU\Taylor';
addpath(genpath(fullfile(repoRoot,'bin')));
cd(fullfile(repoRoot,'example1','FWD3D'));
FWD3D
```

> **Why the working folder matters.** FWD3D calls inpt3 by name and constructs workDir from pwd. Running it from bin will fail unless an inpt3.m is also present there, and outputs will be written wherever pwd points.

Expected result from the supplied input

- compFlag = [1 0 0 1] enables gravity (slot 1) and AEM (slot 4).
- obsData1.dat contains 121 receivers x 7 gravity channels = 847 rows.
- obsData4.dat contains 121 receivers x 4 AEM channels = 484 rows.
- FieldsP1, FieldsP4, ModelP1, ModelP4, and work are created or reused.
- A source-level gravity spot check at X = -250 m, Y = -250 m, Z = -30 m, channel 4 gives 0.030789074 mGal, matching the supplied obsData1.dat.

> **Session side effect.** The first line of FWD3D is clc, clear, close all. Unsaved workspace variables are cleared and open MATLAB figures are closed.

3. Installation and requirements

Required repository content

Item	Purpose
bin/FWD3D.m	Top-level driver script
bin/*.m	Grid, model, forward-solver, and plotting helpers
bin/green3d	AEM Green's-function code and compiled normal/grint MEX binaries
working folder/inpt3.m	Run-specific configuration

MATLAB capabilities

- Gravity and magnetic paths use ordinary MATLAB numerical and graphics functions.
- The AEM path uses parfor in getPredAEM.m; Parallel Computing Toolbox is therefore expected.
- AEM also requires compiled normal and grint MEX binaries compatible with the operating system and architecture. The repository contains Windows 32/64-bit and Linux 64-bit binaries, but no macOS binary.
- The complete bin tree should be added with genpath so that bin/green3d is included.
- The working folder must be writable because data, images, and AEM scratch files are overwritten or created there.

> **Current environment check.** In the reviewed Windows installation, MATLAB resolves normal.mexw64 and grint.mexw64 from bin/green3d. This confirms discovery of the AEM binaries; it is not a full rerun of the expensive AEM example.

4. Coordinates, grids, and property conventions

Coordinate system

Coordinate	Convention	Unit
X	East	m
Y	North	m
Z	Positive downward	m
Surface	Normally Z = 0	m
Airborne receivers	Use negative Z above the surface; the example uses Z = -30	m

The forward and display bounds use the order [xmin xmax ymin ymax zmin zmax]. Cell locations are stored at cell centers. In the supplied model, the anomalous body spans Z = 50 to 100 m and the receivers sit 30 m above the surface.

Grid construction

- xySize{ip} = [dx dy] defines uniform horizontal cell widths.
- A scalar dz{ip} creates uniform vertical cells. The code uses floor((zmax-zmin)/dz), so choose a spacing that divides the vertical interval cleanly.
- A vector dz{ip} defines one thickness per vertical cell. Its cumulative cell centers must remain inside zmax.
- The number of cells is Nx x Ny x Nz. For the supplied anomaly grid, 20 x 20 x 5 = 2,000 cells.
- Grid order follows MATLAB ndgrid: X changes fastest, then Y, then Z. Multi-component model files place all cells for component 1 first, followed by component 2, and so on.

Property conventions

Slot	Model property	Input unit	Image prefix
1 Gravity	Density contrast / density parameter	g/cm3	Dens
2 Magnetic	Scalar magnetic susceptibility	SI	Sus
3 Vector magnetic	Three component property [Mx My Mz]	SI	Mx, My, Mz, Sus
4 AEM	Electrical conductivity for computation	S/m	Resistivity images in Ohm-m

> **AEM unit conversion.** AEM input properties bckg{4} and anom{4} are conductivities. showModel plots 1/conductivity, so barLims{4} and lims3D{4} must be specified in resistivity (Ohm-m).

5. Input file reference

The input is ordinary executable MATLAB code. Variable names and slot numbers are part of the interface and must be preserved.

5.1 Common per-physics model variables

Variable	Meaning and constraints
bckg{ip}	Background property. Scalar for slots 1, 2, and 4; 1x3 row vector for slot 3. Intended layered form is one row per layer.
thick{ip}	Layer thickness vector. Use [] for the reliable half-space path in the distributed version.

Variable	Meaning and constraints
anomBounds{ip}	[xmin xmax ymin ymax zmin zmax] for the forward-model grid.
xySize{ip}	[dx dy] horizontal cell size in meters.
dz{ip}	Scalar uniform thickness or vector of vertical cell thicknesses, in meters.
anom{ip}	Uniform total property, a 1x3 vector for slot 3, [] to use background, or a character-vector filename containing the full model.

A numeric anom value is replicated into every cell inside anomBounds. To describe heterogeneous cells, save the complete total-property model as an ASCII numeric file and assign its path with single quotes, for example anom{1} = 'density_model.dat'. A MATLAB string object is not recognized by the current ischar test.

5.2 Receiver geometry and channel selection

rx{ip}: Receiver X coordinates; vector or array

ry{ip}: Receiver Y coordinates; same number of elements as rx

rz{ip}: Receiver Z coordinates; same number of elements as rx

rc{ip}: Channel codes to calculate; duplicates are removed and codes are sorted

FWD3D flattens the coordinate arrays and removes duplicate [X Y Z] triplets. The supplied 11 by 11 receiver grid produces 121 unique locations. Field-map generation expects the locations to span both X and Y; a line survey or single point can make griddata/contour plotting fail even if the forward calculation itself is valid.

5.3 Physics activation

```
compFlag = [gravity magnetic vectorMagnetic AEM];
compFlag = [1       0        0              1  ];
```

Each nonzero entry enables the corresponding fixed slot. Np is computed as length(bckg), so define the required bckg cells consistently through the highest slot you intend to use.

5.4 Magnetic inducing field

Variable	Meaning	Unit
Bo	Magnitude of the inducing field	nT
Ao	Model/survey azimuth reference	degrees
Io	Field inclination, positive down	degrees
Do	Field declination	degrees

Bo, Ao, Io, and Do are used by both magnetic slots 2 and 3. The code computes direction cosines using Do - Ao under the X-east, Y-north, Z-down convention.

5.5 Display variables

Variable	Meaning
dispLims	Display/interpolation bounds [xmin xmax ymin ymax zmin zmax]. Does not change the forward grid.
dispXY	Display-grid [dx dy] in meters.
dispDz	Display-grid vertical cell size(s) in meters.
barLims{ip}	Two-element color scale [minimum maximum] in plotted property units.
lims3D{ip}	One or more [minimum maximum] ranges selecting cells shown in 3D.
xs, ys, zs	Requested X, Y, and Z slice coordinates. Use [] for no slices or vectors for multiple slices.

6. Worked example: example1/FWD3D/inpt3.m

6.1 Model definition

Part	Supplied value
Gravity	Background 0 g/cm3; uniform body 1 g/cm3; enabled
Magnetic	Background 0 SI; uniform body 0.06 SI; disabled
Vector magnetic	Background [0 0 0]; body [1 0 1] x 0.06/sqrt(2); disabled
AEM	Background 0.001 S/m (1000 Ohm-m); body 0.01 S/m (100 Ohm-m); enabled
All slots	Body grid from -100 to 100 m in X/Y and 50 to 100 m in Z; 10 m cells

6.2 Survey and display

- Receiver grid: X and Y from -250 to 250 m in 50 m steps; Z = -30 m.
- Gravity channels: 4:10, giving Gz and six tensor components.
- AEM channels: 1:4, giving the three coplanar frequencies and the 5.5 kHz coaxial channel.
- Display domain: -300 to 300 m in X/Y, 0 to 200 m in Z, with 10 m display cells.
- Slices: X = 0 m and Z = 75 m; no Y slice.
- 3D density view selects 0.8 to 1.2 g/cm3; AEM 3D view selects 50 to 150 Ohm-m.

6.3 Reusing the example

1. Copy the complete example1/FWD3D folder to a new run folder so earlier outputs remain recoverable.
2. Edit the copied inpt3.m, keeping all four slot indices and required display variables.
3. Change MATLAB's current folder to the new run folder.
4. Add bin recursively to the path and call FWD3D.
5. Check obsDataN.dat row counts and review the field/model PNG files before using the data in inversion.

7. Outputs and file formats

7.1 Top-level outputs

Path pattern	Content
obsData1.dat	Gravity synthetic data for slot 1
obsData2.dat	Induced magnetic synthetic data for slot 2
obsData3.dat	Vector magnetic synthetic data for slot 3
obsData4.dat	Complex AEM synthetic data for slot 4
FieldsPN	One field-map PNG for every selected channel
ModelPN	Requested slices and 3D model images
work	AEM-only per-channel solver scratch folders

> **Overwrite behavior.** Repeated runs reuse these names. MATLAB save and saveas replace matching result files, while getPredAEM deletes matching fwd*, sig*, int*, and recp* files inside each active AEM work subfolder before solving.

7.2 Real-valued data schema

```
X[m]   Y[m]   Z[m]   channel_code   synthetic_value
```

This five-column schema applies to gravity and both magnetic modes. Gravity and magnetic rows are ordered by receiver with channel changing fastest, but downstream code should use the explicit channel column instead of relying on row order.

7.3 Complex AEM data schema

```
X[m]   Y[m]   Z[m]   channel_code   real_ppm   imaginary_ppm
```

AEM rows are grouped by channel and then by receiver coordinates. Real and imaginary values are separate ASCII columns. The top-level file is named obsData4.dat even though it contains synthetic predictions; this naming is intentional because INV3D consumes it as observed data.

7.4 AEM work folders

getPredAEM creates work/01, work/02, and so on, one folder for each selected channel position. The folder number is the position within rc{4}, not necessarily the channel code. For rc{4} = [1 4], the folders are 01 and 02.

Typical file	Purpose
intem3d.par	Generated INTEM3DQL parameter script
recpar.dat	Receiver/source/frequency requests
sigbody.dat	Total conductivity model
sigbody_tmp.dat	Solver working copy
inputpara.mat	Parsed solver input parameters

8. Visualization controls

8.1 Model slices and 3D views

showModel interpolates the forward model into the display domain, applies the selected color scale, then writes axis slices and up to three orientations of the selected 3D cells. Slice file names end in -XsliceN, -YsliceN, or -ZsliceN. 3D views end in -3D, -3Dside, and -3Dtop.

8.2 Choosing useful view ranges

- barLims changes the color mapping only; it does not clip or alter the forward model.
- lims3D selects cells for the 3D render. Multiple rows are treated as a union of accepted property ranges.
- Outside anomBounds, the display model is filled with the background property.
- A requested slice is drawn when its coordinate lies within half a cell of a display-grid center.

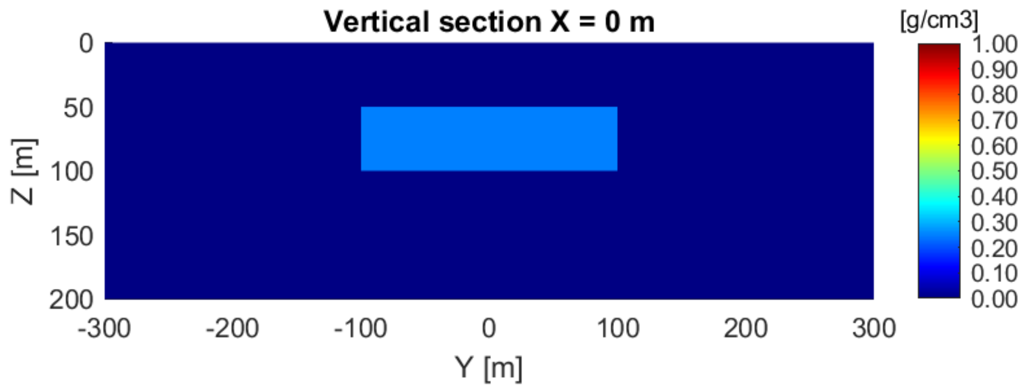

Figure 1. Supplied gravity-density X slice. Z increases downward and the body occupies 50-100 m depth.

8.3 Field maps

showFields interpolates each channel onto a 65 by 65 map grid using griddata and writes a contour plot. FWD3D passes the synthetic values in the columns labeled observed by the plotting helper, so the panels are titled Real observed and, for AEM, Imag observed.

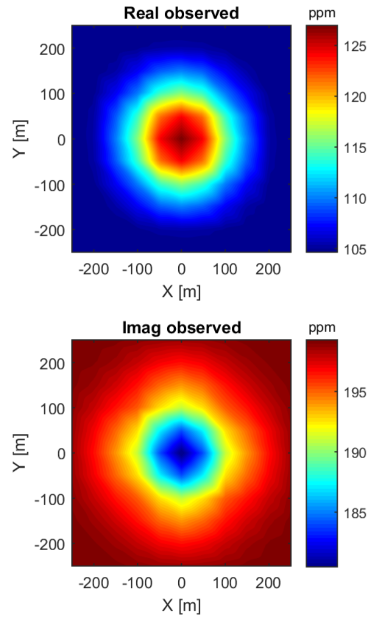

Figure 2. Supplied AEM channel-1 map, showing real and imaginary synthetic response in ppm.

> **Interpreting the labels.** Within a forward run, observed in these figures means the value stored as the synthetic observation. Predicted panels appear only when showFields receives an additional prediction column, as it does in inversion-result plotting.

9. Building a new input file

Recommended sequence

1. Choose the enabled physics slots and set compFlag.
2. Define background property, half-space thickness setting, model bounds, and cell sizes for every slot.
3. Define the total anomaly property. Use a scalar/vector for a uniform block or a character-vector file path for a heterogeneous model.
4. Build matching rx, ry, and rz arrays, then choose valid rc channel codes.
5. For magnetic calculations, set Bo, Ao, Io, and Do.
6. Choose a display domain that contains the anomaly and requested slices.
7. Set barLims in plotted units and lims3D to isolate the cells of interest.
8. Run from a dedicated writable folder and inspect both numeric and image outputs.

Heterogeneous model files

The file must contain exactly Ncells x Ncomponents numeric values. FWD3D loads the file, reshapes it to the background-model shape, and treats the values as total property rather than anomaly contrast. Use ASCII numeric data with no header.

```
% Example: use a file for a heterogeneous gravity model
anom{1} = 'density_total.dat';
% File order: X fastest, then Y, then Z
```

> **Do not pass a full numeric cell vector directly.** A nonempty numeric anom value is replicated once per cell. Use a filename for a spatially varying model.

Pre-run checklist

- pwd is the intended run folder and it contains inpt3.m.
- which FWD3D and which getPredAEM resolve to the intended bin tree.
- For AEM, which normal and which grint resolve to compatible MEX binaries.
- rx, ry, and rz contain equal numbers of elements and span a 2D area if maps are required.
- Bounds and cell sizes produce at least two display cells in X, Y, and Z for interpolation.
- barLims has a nonzero range and lims3D uses the plotted property units.
- Any existing outputs have been copied elsewhere if they must be retained.

10. Performance and troubleshooting

10.1 Performance guidance

- Gravity and magnetic work scales approximately with receivers x channels x model cells.
- AEM is substantially more expensive because it runs INTEM3DQL separately for each selected channel and uses large scratch matrices.
- Start with a coarse grid and one or two receiver/channel combinations before expanding a new AEM survey.
- Keep anomBounds tight around the region that needs 3D cells. Use the larger dispLims only for presentation.
- Reduce rc to the channels actually required. AEM work is parallelized across the selected channel positions.
- Keep work on a fast local disk and ensure enough free space for MAT and ASCII scratch files.

10.2 Troubleshooting

Symptom	Likely remedy
Undefined function FWD3D or helper	Use addpath(genpath(fullfile(repoRoot,'bin'))) and confirm which FWD3D.
Undefined inpt3	Change to the run folder containing inpt3.m, or place the intended file on the MATLAB path.
AEM cannot find normal or grint	Confirm bin/green3d is on the path and that the MEX extension matches the OS and architecture.
AEM parfor/license error	Verify Parallel Computing Toolbox and an available license.
griddata/contour error	Use receivers spanning both X and Y, or temporarily disable the showFields call for line/single-point testing.
Empty or missing 3D image	Adjust lims3D so at least one displayed cell falls within a selected range.
Misleading/flat colors	Set barLims to a nonzero range that brackets the plotted values; use resistivity for slot 4.
Interpolation error in showModel	Ensure forward and display grids contain more than one cell in X, Y, and Z.
Unexpected files or overwritten output	Run in a dedicated folder and archive prior obsData, FieldsP, ModelP, and work folders.

10.3 Known limitations in the distributed source

- The input filename in FWD3D.m is hard-coded as inpt3. Renaming the configuration requires editing the script or providing a wrapper/copy.
- The four physics meanings are fixed by cell index. Additional or reordered physics are not supported by configuration alone.
- The layered-background branch is not reliable as distributed: getMpars.m references an undefined variable Nc in the last-layer case, and interpM.m references a depth field that it does not create. Use thick{ip} = [] unless those code paths are corrected and tested.
- AEM frequencies, transmitter/receiver offsets, and coplanar/coaxial orientations are hard-coded in getPredAEM.m; inpt3.m can select channels but cannot redefine the system.
- AEM depends on legacy INTEM3DQL code and precompiled Green's-function MEX files. Portability to unsupported operating systems requires recompilation and validation.
- Output plotting is unconditional. A valid numerical run can still stop during visualization if the receiver geometry or display limits are unsuitable.
- The supplied example enables only gravity and AEM. Magnetic slots are configured but not represented by supplied obsData2/obsData3 artifacts.

> **Scope of this manual.** These limitations describe the repository version reviewed for this manual. If helper code is revised, revalidate the affected behavior and update the manual alongside the code.

Appendix A. Channel reference

A.1 Gravity channels

Code	Component	Output unit
1	Gt	mGal
2	Gx	mGal
3	Gy	mGal
4	Gz	mGal
5	Gxx	Eotvos
6	Gyy	Eotvos
7	Gzz	Eotvos
8	Gxy	Eotvos
9	Gzx	Eotvos
10	Gzy	Eotvos
11	Gd = 0.5(Gxx-Gyy)	Eotvos

A.2 Magnetic channels

Code	Component	Output unit
1	TMI	nT
2	Hx	nT
3	Hy	nT
4	Hz	nT

The same codes are used by scalar induced magnetics (slot 2) and the vector magnetic parameterization (slot 3).

A.3 DIGHEM-style AEM channels

Code	Geometry	Frequency	Output
1	Coplanar	56,000 Hz	ppm, real + imaginary
2	Coplanar	7,200 Hz	ppm, real + imaginary
3	Coplanar	900 Hz	ppm, real + imaginary

Code	Geometry	Frequency	Output
4	Coaxial	5,500 Hz	ppm, real + imaginary
5	Coaxial	900 Hz	ppm, real + imaginary

The supplied rc{4} = 1:4 selects the first four channels. Channel 5 is implemented even though it is not selected in the example.

Appendix B. Annotated sample input

The following compact listing preserves the important assignments from example1/FWD3D/inpt3.m while adding explanatory comments.

```
% Slot 1: gravity density model
bckg{1}=0;  thick{1}=[];
anomBounds{1}=[-100 100 -100 100 50 100];
xySize{1}=[10 10];  dz{1}=10;  anom{1}=1;

% Slot 2: scalar magnetic susceptibility model
bckg{2}=0;  thick{2}=[];
anomBounds{2}=[-100 100 -100 100 50 100];
xySize{2}=[10 10];  dz{2}=10;  anom{2}=0.06;
Bo=50000;  Ao=0;  Io=75;  Do=25;

% Slot 3: three-component magnetic model
bckg{3}=[0 0 0];  thick{3}=[];
anomBounds{3}=[-100 100 -100 100 50 100];
xySize{3}=[10 10];  dz{3}=10;
anom{3}=[1 0 1]*0.06/sqrt(2);

% Slot 4: conductivity model (S/m)
bckg{4}=1/1000;  thick{4}=[];
anomBounds{4}=[-100 100 -100 100 50 100];
xySize{4}=[10 10];  dz{4}=10;  anom{4}=1/100;

% Shared 11 x 11 receiver grid at 30 m elevation
rx{1}=repmat(-250:50:250,1,11)';
ry{1}=repmat(-250:50:250,11,1); ry{1}=ry{1}(:);
rz{1}=zeros(size(rx{1}))-30;  rc{1}=4:10;

% Slots 2-4 use the same coordinate construction
% rc{2}=1:4; rc{3}=1:4; rc{4}=1:4;

% Enable gravity and AEM only
compFlag=[1 0 0 1];

% Display grid and slices
dispLims=[-300 300 -300 300 0 200];
dispXY=[10 10];  dispDz=10;
xs=0;  ys=[];  zs=75;

% Color and 3D selection ranges
lims3D{1}=[0.8 1.2];     barLims{1}=[0 1];
lims3D{2}=[0.05 0.07];   barLims{2}=[0 0.06];
lims3D{3}=[0.05 0.07];   barLims{3}=[-0.06 0.06];
% Slot 4 ranges are resistivity (Ohm-m), not conductivity
lims3D{4}=[50 150];      barLims{4}=[1 1e3];
```